Vaibhav Krishan

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Research Positions

PostDoc (CSE)	Indian Institute of Technology Madras	Aug 2026-
PDF (TCS)	The Institute of Mathematical Sciences	July 2024-July 2026

Education

Ph.D. Computer Science and Engineering	Indian Institute of Technology Bombay Advised by Prof. Sundar Vishwanathan and Prof. Nutan Limaye, CPI: 9.09/10	July 2017-May 2024
B.Tech. Computer Science and Engineering	Indian Institute of Technology Bombay CPI: 7.59/10	July 2009-May 2013

Journal Publications

Algorithmica
A #SAT Algorithm for Small Constant-depth Circuits with PTF gates
with S. Bajpai, D. Kush, N. Limaye and S. Srinivasan
Algorithmica 84, 1132-1162 (2022).

Conference Publications

FSTTCS 2026
On Knowledge Compilation Languages for QBFs
with S. Chede and A. Shukla
The 46th IARCS Annual Conference on Foundations of Software Technology and Theoretical Computer Science, 2026 (to appear).

MFCS 2026
On CC^0 Lower Bounds for AND via Torus Polynomials
with J. Sarma
The 51st International Symposium on Mathematical Foundations of Computer Science, 2026.

SAT 2026
On Proof Systems for #QBF
with S. Chede, L. Chew and A. Shukla
The 29th International Conference on Theory and Applications of Satisfiability Testing, 2026.

ITCS 2026
Lower Bounds and Separations for Torus Polynomials
with S. Vishwanathan
The 17th Innovations in Theoretical Computer Science Conference, 2026.

CSR 2021
Upper Bound for Torus Polynomials
The 16th International Computer Science Symposium in Russia, 2021.

ITCS 2019	A #SAT Algorithm for Small Constant-depth Circuits with PTF gates with S. Bajpai, D. Kush, N. Limaye and S. Srinivasan The 10th Innovations in Theoretical Computer Science Conference, 2019.

Preprints

ECCC	Isolation Lemma for Directed Reachability and NL vs. L with N. Limaye

Talks and Presentations

Presentation	Lower Bounds and Separations for Torus Polynomials ITCS 2026.
Video	Lower Bounds and Separations for Torus Polynomials ITCS 2026.
Lecture	Polynomials and Computation Foundational Lecture Series on TCS, IMSc, 2025.
Presentation	Upper Bound for Torus Polynomials CSR 2021. (online)
Presentation and Poster	A #SAT Algorithm for Small Constant-depth Circuits with PTF gates ITCS 2019.

Service and Engagement

Sub-reviewer	SAT 2026.
Sub-reviewer	Computational Complexity Conference (CCC) 2024.
Sub-reviewer	Foundations of Software Technology and Theoretical Computer Science (FSTTCS) 2024
Volunteer	FSTTCS 2019

Teaching Assistance

- Awarded best TA of the semester for CS 101.
- Awarded best TA of the month for Automata Theory.
- TA duty for complexity theory twice, design and analysis of algorithms, combinatorics, and theoretical machine learning.

Professional Experience

- Freelance work in quantitative trading for multiple clients since May ‘22.
- Quantitative Research Consultant for AlphaGrep, and for 5-Swans, during official break from PhD June ‘21-May ‘22.
- As Quantitative Developer in AlphaGrep from June ‘16-July 17 and Jan ‘14-April ‘15.
- As Data Scientist in Housing.com from May ‘15-May ‘16.
- As Software Engineer in Samsung from July ‘13-Dec ‘13.

Personal Details

D.O.B.	09th December 1993
Sex	Male
Nationality	Indian